

Term: Fall 2023

Subject: Maths (MAT)

Course Number: 142

Assignment 1 (20 points)

Due on Friday, September 8th at 11:59 pm

Q1. The roots of the quadratic equation $x^2 - 5x + 6 = 0$ are:

- 1) (1, 6)
- 2) (2, 3)
- 3) (3, 5)
- 4) (-2, -3)

Q2. The simple interest on ₹5000 at 10% per annum for 2 years is:

- 1) ₹1000
- 2) ₹1050
- 3) ₹1100
- 4) ₹1200

Q3. The derivative of $\ln(x^2+1)$ with respect to x is:

- 1) $1/x$
- 2) $2x/(x^2+1)$
- 3) $1/(x^2+1)$
- 4) $2x$

Q4. The exact value of $\cos(2\theta)$ is:

- 1) $1 - 2\sin^2\theta$
- 2) $\sin^2\theta + \cos^2\theta$
- 3) $2\sin\theta\cos\theta$
- 4) $\tan\theta$

Q5. A fair die is rolled. The probability of getting a number greater than 4 is:

- 1) $1/6$
- 2) $2/6$
- 3) $3/6$
- 4) $4/6$

Q6. The determinant of the identity matrix of order 3 is:

- 1) 0
- 2) 1
- 3) 2
- 4) 3

Q7. $\lim_{x \rightarrow 0} (1 - \cos x) / x^2 =$

- 1) 0
- 2) $1/2$
- 3) 1
- 4) ∞

Q8. If a sum of ₹10,000 amounts to ₹12,100 in 2 years at compound interest, the rate per annum is:

- 1) 5%
- 2) 10%
- 3) 12%
- 4) 15%

Q9. $\sin^2\theta + \cos^2\theta =$

- 1) 0
- 2) 1
- 3) $\sin 2\theta$
- 4) $\cos 2\theta$

Q10. The sum of the first n terms of an arithmetic progression with first term a and common difference d is:

- 1) $n/2 [2a + (n-1)d]$
- 2) a^n
- 3) $n(a+d)$
- 4) $(a+d)^n$